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Review

A review of energy storage technologies for demand-side management in industrial facilities

Joseph Elio [a], Patrick Phelan [a], Rene Villalobos [b], Ryan J. Milcarek [a,*]

[a] School for Engineering of Matter, Transport and Energy, Arizona State University, 501 E. Tyler Mall, Tempe, AZ, 85287-6106, USA
[b] School of Computing, Informatics, and Decision Systems Engineering, Arizona State University, Tempe, AZ, USA

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ABSTRACT

Demand-side management (DSM) in industrial facilities provides an opportunity for substantial amounts of energy cost savings, since industrial facilities are the largest energy consuming sectors globally. In this work, energy storage (ES) technologies are critically reviewed and compared with industrial DSM in mind. ES technologies reviewed herein include lithium-ion battery energy storage (BES), sodium-sulfur BES, lead-acid BES, flow BES, supercapacitor ES, superconducting magnetic ES, thermal ES, flywheel ES, pumped hydro ES, and compressed air ES. The fundamentals of these energy storage technologies are reviewed in detail including recent developments, followed by case studies and extensive comparisons. These comparisons include, but are not limited to cost per cycle analyses, levelized cost of energy analyses, and comparisons between performance, transient, and cost characteristics. Some key properties analyzed include the rated power, power density, efficiency, lifetime, discharge time, capital costs, and O&M costs.

1. Introduction

Progress in electric-powered technologies is causing rapid development of transportation, communication, industry, and other functions that are improving humanity (Pavlov, 2017). As the world's population continues to climb, so has its need for the use of consumable energy sources. Energy demand is projected to more than double by 2050 and triple by 2100 (Guney and Tepe, 2017). This rising consumption will inevitably produce greater environmental concerns such as carbon emissions and the impact on global climate. The need for renewable energy systems (RESs) has resulted in an increased interest in energy storage (ES) technologies to mitigate the stochasticity of renewable energy sources. For example, RESs are steadily increasing their contribution to global energy production: from 18.1% in 2017 to 26% in 2019 (Mostafa et al., 2020). In the same way, ES is steadily rising in capacity, predicted to grow from 1.4 GW and 2.3 GWh in 2017 to 8.6 GW/21.6 GWh in 2022 (Mostafa et al., 2020). Power generation plants are second only to transportation as the largest contributor to global carbon emissions (Environmental Protection Agency, 2019), because they must be continually ready to meet sudden and unpredictable loads. It is crucial to develop a solution that will effectively reduce demand on these plants. Industrial facilities are considered to be the leading users of energy at 54% of the world's total delivered energy (Haiwei and Wang, 2009). Therefore, ES technologies should be matched to a facility to reduce or shift maximum power demands away from the power plant, a process known as demand-side management (DSM).

Storing energy from a supply (power plants or RESs) for the highest consumers (industrial facilities) will reduce harm to the environment and diminish energy costs because this stored energy is then discharged to shift peak loads from power generation plants. This is an important opportunity because the additional strain on the power grid caused by these facilities forces utility companies to charge more during their grid's peak loads to avoid power outages, a cost passed on to the consumers. By increasing industrial facilities' usage of ES technologies, utility companies could expect to provide a 'smooth' aggregate load– a transition from the current 'peak' and 'valley' loads. Through the approaches of 'peak shaving' (discharging) and 'valley filling' (charging) an ES system can smooth the aggregate load of the power grid. As a result, the grid becomes more efficient and cost effective to run because it is less dependent on spinning reserve power generation plants (Lim et al., 2020). Industrial facilities have tremendous potential to decrease their energy and demand costs through means of ES to shave the peak load off the power grid, bringing greater balance between production and demand, while simultaneously improving the reliability and

* Corresponding author.
E-mail addresses: Rene.Villalobos@asu.edu (R. Villalobos), Ryan.Milcarek@asu.edu (R.J. Milcarek).

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financial performance of the power grid (Tronchin et al., 2018).

Other review papers have been written on the topic of DSM and/or ES devices. For example, Tronchin et al. (2018) focused on DSM from a multi-level energy modelling strategy and briefly mentioned ES devices and their respective levelized costs. Furthermore, Koohi-Fayegh et al. (Koohi-Fayegh and Rosen, 2020) produced an in-depth analysis of ES types, applications, and recent developments (2020), without conducting case studies or clear comparisons between technologies. Hall & Bain (Hall and Bain, 2008) reviewed ES technologies, primarily focusing on electrical and electrochemical ES, also mentioning superconducting magnetic energy storage and flywheel energy storage. Luo et al. (2015) also presented a comprehensive review of ES technologies, again without conducting any case studies and having few comparisons between technologies. Gao (2015) wrote about ES for microgrids, focusing on ES technologies and control methods. Zablocki (2019) presented a brief fact sheet about current ES technologies. Zakeri & Syri (Zakeri and Syri, 2015) compared life cycle costs of ES systems. Similarly, Chen et al. (2009) reviewed electrical ES systems and compared the respective technical characteristics. None of these review papers, however, provides a thorough review, comparison, and application of ES technologies for DSM, especially in the context of industrial facilities.

This work provides a guide for ES technologies to be used for industrial DSM. Promising ES technologies are presented and compared in this work to guide industrial facilities to an ES technology that meets the facility's needs. There is a lack of research reviewing ES for use as part of an industrial DSM strategy. At the same time, opportunities for industrial DSM are significant, as over half of global power generation is spent in these facilities. This work demonstrates, through case studies, that industrial facilities are unique when compared to other facilities. Specifically, since industrial facilities are much larger, different technologies can be used and electrical energy is generally stored at a far greater capacity than other facilities (e.g., commercial, institutional, public), which typically store thermal energy. The list of DSM technologies is organized depending upon their respective power capacities, performance parameters, transient properties, and costs. Some key parameters analyzed include the rated power, power density, efficiency, lifetime, discharge time, capital costs, and O&M costs. Lastly, case studies of ES technologies applied to industrial facilities, commercial facilities, educational institutions, and public buildings are analyzed based on their frequency of installation and power capacity.

2. Methods

This review is unique due to its focus on industrial facilities, with a particular focus on verifying "well-rounded" energy storage technologies for use in industrial DSM. In the creation of this review, the following research method was utilized: 1) general review papers pertaining to energy storage were reviewed and analyzed to determine energy storage technologies suitable for industrial facilities; 2) Numerous recent publications were found by searching keywords on ScienceDirect, Scopus, and Google Scholar databases to find peer-reviewed journal papers and textbook chapters indicating the current state-of-the-art of each selected technology. The literature was reviewed manually by first reading the abstract and results to determine if the information represented novel advancements in the technology's state-of-the-art and could provide a useful contribution to this work. Once the research was conducted, the paper was written in the following sequence to narrow the vast list of technologies to only include "well-rounded" technologies: 1) technology sections were written by first highlighting background information about the governing physics, 2) followed by advantages of the technology, 3) disadvantages of the technology, and 4) case studies of the technology used in industrial facilities when applicable. After each technology was introduced, performance values from multiple sources were tabulated and analyzed to compare the technologies. By taking the average midpoint in the full collection of reported ranges, nominal performance parameters are derived and analyzed along with a capital cost per cycle metric and levelized cost of energy metric to determine "well-rounded" energy storage technologies for use in industrial DSM. The database (Sandia National Laboratories, 2020) provided by Sandia National Laboratories was selected to analyze case studies because it is a comprehensive collection of world-wide energy storage applications.

3. Energy storage technologies

As previously mentioned, global energy consumption is steadily rising, some of which is due to inefficient peaker plants ramping to accommodate industrial facilities and their increased consumption. A promising method to reduce a facility's electricity demand is to discharge an ES device during peak hours and charge the device during off-peak hours, known as 'peak shaving' and 'valley filling'. The energy can be stored in different forms including electrical, electrochemical, magnetic, thermal, and mechanical. An overview of ES technologies discussed in this work, sorted by the form of energy storage, is presented in Fig. 1. Each technology is reviewed in detail: fundamentals, characteristics that make it suitable or not suitable for industrial DSM, and case studies of its current application for DSM in facilities (see Appendix A).

Key performance parameters of ES devices include, but are not limited to rated power, specific energy, energy density, specific power, power density, charge time, storage time, response time, discharge time, round trip efficiency, depth of discharge and the lifetime of the device. The rated power is the maximum power capacity of the device, that is, the most power a device can render in a given period of time. Energy and power are presented in both 'per unit mass' (specific energy and specific power) and 'per unit volume' (energy density and power density) basis in order to compare both by the mass and/or volume of the technology. Depending on the application and industrial facility, restrictions on size or weight can be a factor in the choice of ES. Charging and storage times are the length of time required to charge the technology and the time that charge will be held in the device, respectively, without experiencing significant self-discharge effects. Response time is the elapsed time between requesting power from the device and the device sending enough power to meet the load. Discharge time is the duration that the device can supply power to the load. Round-trip efficiency, or storage efficiency, noted as 'efficiency' in this work, is the ratio of total energy extracted from the device in discharge to the total energy that was supplied to the device in charging. Depth of discharge is the useable storage capacity of the device (Thien et al., 2015). For example, most batteries that are deeply discharged will experience a negative effect on their lifetime duration, or the number of charge/discharge cycles it can maintain. Conversely, mechanical systems do not have an electrochemical constraint on their depth of discharge, but a geometric constraint on the storage device. This is demonstrated when looking at the geometry of a compressed air receiver or the upper reservoir of a pumped hydro system. If the upper reservoir of a pumped hydro system is optimally designed, all the stored water can be sent to the lower reservoir and the system will achieve 100% depth of discharge.

3.1. Overview of Energy Storage Technologies

In this section, ES technology key performance parameters are tabulated for ease of viewing in Table 1, including the rated power capacity, specific energy, energy density, specific power, power density, round-trip efficiency, suitable depth of discharge, and lifetime. In Table 2, the charge time, suitable storage duration, response time, and discharge time are displayed for the ES devices reviewed above. TES is neglected from this list, as it is not as relevant to some industrial facilities and is likely used as a supplement to another form of ES listed below.

Table 1 is used to analyze key performance parameters against one another. This is imperative in an industrial facility's search for a fitting ES technology. For example, the high energy density, power density, and

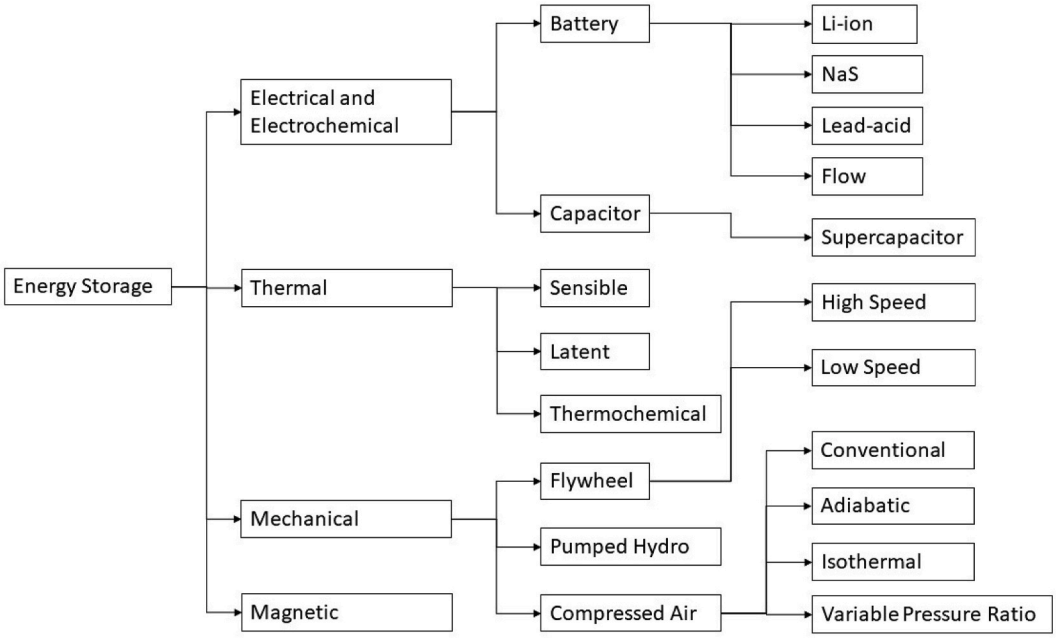


Fig. 1. Overview of energy storage technologies.

Table 1
Performance metrics of energy storage devices.

	Power Capacity (MW)	Specific Energy (Wh/kg)	Energy Density (Wh/L)	Specific Power (W/kg)	Power Density (W/L)	Round-Trip Efficiency (%)	Suitable Depth of Discharge (%)	Lifetime (cycles)
Li-ion BES	0–100	75–270	250–750	150-3000	1500–10,000	90–95	80(1)	2000–10,000
NaS BES	0–100	100–240	150–250	100–230	120–180	75–85	90	2500–4500
Pb-acid BES	0–40	30–50	50–80	75–300	10–400	70–90	60	500-2000
PSB Flow Battery BES	–	15–30	20–30	–	<2	65–85	100	2000–2500
VRB Flow BES	0.03–3	10–30	20–70	50–140	0.5–2	65–85	100 (Lovell, 2019)	10,000–13,000
ZnBr Flow BES	0.05–2	30–50	30–60	50–150	<25	70–80	100	2000–10,000
SC ES	0–0.3	2.5–15	10–30	500-5000	40,000–120,000	90–95	100	100,000–1,000,000
SMES	0.1–10	0.5–5	0.2–2.5	500-2000	1000–4000	95–98	100	>100,000
FES	0–1.5	10–100	20–80	400-1500	1000–5000	93–95	100	20,000–100,000
PHES	100-5000	0.5–1.5	0.5–1.5	0.5–1.5	0.5–1.5	75–85	≤100	20,000–50,000
CAES	5–400	30–60	2–6	–	0.2–0.6	40–70	≤100	8000–12,000

Table 2
Time dependence of energy storage devices.

Technology	Charge Time	Suitable Storage Duration	Response Time	Discharge Time
Li-ion BES	2–3 h	min-days	~ms	min-h
NaS BES	–	s-h	<10s	s-h
Pb-acid BES	8–10 h	min-days	~ms	s-h
PSB Flow BES	–	h-months	<100 ms	s-10 h
VRB Flow BES	–	h-months	<100 ms	s-10 h
ZnBr Flow BES	–	h-months	<100 ms	s-10 h
SC ES	1–10s	s-h	~ms	ms-1h
SMES	1–10s	min-h	~ms	ms-8s
FES	1–2min	s-min	~s	ms-15min
PHES	–	h-months	mins	1–24 h+
CAES	8–40 h	h-months	9–12min	1–24 h+

efficiency of FES and SC ES may be deterred by their low rated power. Also, consider the high rated power of PHES is less attractive because of the low energy density and a facility's lack of space for the technology.

As another example, the high power density and efficiency of Li-ion BES could be deterred by the few cycles that it can endure. It must be noted that Table 1 displays the common numbers and ranges found in the full data selection found in the appendix Table B1.

Table 2 is also a necessity when comparing ES technologies, since the transient characteristics are equally important to the performance parameters shown in Table 1. For example, the fast response time of Li-ion BES is complimented by the long discharge time, whereas the fast response time of SC ES is deterred by a short discharge time. In other words, for SC ES only a small amount of energy can be stored resulting in a short discharge time. Neither Table 1 or Table 2 can stand alone as a complete comparison of ES devices, as they must be used together to ensure all characteristics of the ES technologies are considered. In section 4, comparisons using values from Tables 1 and 2 are used in conjunction with cost parameters presented in section 3.2 to determine the most promising candidates for ES in industrial facilities. It should be noted that Tables 1 and 2 display the common ranges found from the full data selection reported in the appendix Table B1 and Table B2, respectively.

3.2. Costs of ES technologies

In this section, all ES devices will have their costs compared against one another. Note that all values are represented in United States currency ($). "Capital Costs" include all equipment and installation costs (in $/kW) for balance of plant (power transformer, protective devices, cooling system (Mostafa et al., 2020)) and power conversion systems. "Capital Storage Costs" includes the costs of the storage container, as well as the construction and commissioning costs (Mongird et al., 2019). Furthermore, operation and maintenance (O&M) costs are presented, where fixed O&M, usually presented in units of $/kW/yr, does not depend on the operation phase of the ES device, whereas variable O&M ($/MWh) does. Total O&M costs are derived by the summation of fixed and variable O&M charges.

Table 3 displays a summary of the full data table (located in appendix Table B3) consisting of capital costs ($/kW), capital storage costs ($/kWh), fixed O&M, variable O&M, and total O&M. These are important metrics to be considered when pairing an ES technology to an industrial facility. These values are further analyzed in sections 3.2.1 and 3.2.2 to derive a capital cost per cycle and levelized cost of energy.

3.2.1. Capital cost per cycle

While Tables 1–3 provide the performance, time, and cost characteristics of each ES technology, it is useful to develop metrics that compare several of the parameters simultaneously. This provides a more complete picture of each individual technology while also providing additional tools for comparison across technologies. In this section a capital cost per cycle is defined to show an investor the up-front cost of an ES device to perform one charge/discharge cycle. Nominal table values are derived from Table B1, Table B2, and Table B3 by neglecting outliers, taking the average minimum range value, average maximum range value, and the midpoint between these values. This procedure essentially yields an "average of averages," to best give a single value for each performance parameter. Based on these nominal values, a capital cost per cycle, *CPC*, is derived using Equation (1):

$$CPC = \frac{CC \times PC + CSC \times PC \times DT}{\#cycles} \tag{1}$$

where *CC* is the capital costs ($/MW), *PC* the power capacity (MW), *CSC* the capital storage costs ($/MWh), *DT* the discharge time (h), and *# cycles* the lifetime (cycles). The devices are compared at fixed power and ES capacities, as shown in Table 4. These power and ES capacities are based on typical power capacities for an industrial facility, selected from the case studies conducted in Section 5. The ES capacity was derived from the power capacity and average discharge time of 4 h for an industrial facility, covering the typical daily 4 h of peak energy costs. However, since SC ES and FES have very low discharge time capabilities (less than 4 h), a power and ES capacity of 100 kW/100 kWh is included in these fixed power and ES capacities so these devices can be compared.

For the capacities 100 kW/100 kWh, SC ES has the lowest capital cost per cycle, but for all other capacities PHES has the lowest capital cost per cycle. This is one reason that PHES is installed to the highest capacity globally. It should also be noted that the capital cost per cycle for Li-ion BES is the highest for every capacity shown, except 100kW/100 kWh. The capital cost per cycle of all ES devices is significant, and this is not including the O&M or cost of energy charges. Although the capital cost of ES is high, the inherent demand savings obtained from peak shaving and valley filling makes ES a valuable asset for many facilities.

Table 4
Capital cost per cycle of ES devices (USD/cycle).

	100 kW/ 100 kWh	100 kW/ 400 kWh	500 kW/2 MWh	1 MW/4 MWh	10 MW/40 MWh
Li-ion	93.71	204.63	1023.14	2046.29	20,462.85
NaS	95.82	139.33	696.67	1393.34	13,933.42
Pb-acid	102.20	175.83	879.16	1758.31	17,583.10
Flow	44.37	73.27	366.36	732.72	7327.17
SC	0.37	–	–	–	–
SMES	7.31	27.88	–	–	–
FES	0.73	–	–	–	–
PHES	9.32	10.18	50.88	101.77	1017.68
CAES	9.05	10.83	54.15	108.30	1082.96

Note '-' indicates that the device cannot meet the power and/or energy storage capacity.

3.2.2. Levelized cost of energy

The levelized cost of energy (LCOE) is typically the most important cost characteristic to an investor because it accounts for the time value of money to accurately portray the cost of each unit of energy extracted from the ES technology. LCOE represents the total cost of a system throughout its lifetime divided by the total energy extracted from the system, as seen in Eqs. (2) and (3):

$$P = \sum CC + \sum_{t=1}^{T} \frac{p_0(1+i)^t}{(1+r)^t} + \sum_{t=1}^{T} \frac{O\&M_0(1+o)^t}{(1+r)^t} \tag{2}$$

$$LCOE = \frac{P\frac{r(1+r)^T}{(1+r)^T-1}}{E_{out\,t}} \tag{3}$$

where P is the total price of the system throughout its lifetime, CC represents the capital costs of the ES device, p_0 the price of energy purchased from the grid in the first year of operation, i the price increase factor of energy, r the discount rate of the investment, $O\&M_0$ the annual operating and maintenance costs for the first operational year, while o is the increase in O&M costs as time progresses, and E_{out} the energy extracted from the device. There is no discount rate applied to the capital costs because it is assumed that all capital costs are paid upfront. It is also assumed that i, r, and o remain constant throughout the lifetime of each ES device, and the device is cycled an equal number of times every year, equivalent to the total life cycles divided by the total lifetime (T, years). The *LCOE* formulation standardizes the different lifetimes of the respective devices with a capital recovery factor.

Table 3
Costs of energy storage technologies.

	Capital Costs ($/kW)	Capital Storage Costs ($/kWh)	Fixed O&M ($/kW/yr)	Variable O&M ($/MWh)	Total O&M ($/MWh)
Li-ion Battery	1200–4000	600-2500	2.2–15.5	0.45–6.3	0.70–8.07
NaS BES	980-2800	300–500	2.24–19.38	0.336–6.272	0.59–8.48
Pb-acid BES	218-3644	206–949	3.6–14.6	0.17–0.58	0.24–2.25
PSB Flow BES	1038–1465	1200–1291			
VRB Flow BES	1430–1847	140-1400	3.81–19.38	0.22–3.14	0.65–5.35
ZnBr Flow BES	700-2520	140-1000	3.58–7.73	0.34–2.24	0.75–3.12
SC ES	140–560	300-5600			
SMES	140–636	980-10,000			
FES	294-2880	2072–28,055	4.8–6.7	0.22–4.3	0.77–5.06
PHES	1700–5100	5–100	2.3–10.4	0.21–0.95	0.47–2.14
CAES	400–800	2–50	2.5–4.2	2.1–3.4	2.39–3.88

The projected retail price of electricity in 2021, for the industrial sector, is taken to be 7.01 cents per kWh (U.S. Energy Information Administration (EIA), 2020), or $70.10/MWh. The value of i will be taken as 3.9%, a reasonable estimate of the increase rate in the cost of energy, and the increase rate from 6.75 to 7.01 cents per kWh from 2020 to 2021 (U.S. Energy Information Administration (EIA), 2020). The discount rate, or price increase rate, can be taken as high as 6% (Pawel, 2014), so 5% will be taken as the value of r for a conservative estimate. The value of o is estimated as 3% based on the value reported for one facility (National Science Board, 2018). Using midpoint values from the lifetime ranges reported in (Chen et al., 2009), the LCOE results are shown below in Table 5 for the previously defined power and ES capacities of industrial facilities.

From Table 5, it can be observed that SC ES has the lowest LCOE for 100 kW/100 kWh, SMES has the lowest for 100 kW/400 kWh, and flow BES has the lowest in all other capacities. Another useful way to analyze the LCOE is through plots, where the power capacity is a fixed value, standard for an industrial facility, and the energy storage capacity is varied, as seen in Fig. 2.

When analyzing these plots, note that plots A) and B) represent a power capacity of 100 kW and 500 kW respectively, where SC, SMES, and FES can supply this amount of power, but not for a long duration (up to approximately 100 kWh). In contrast, Fig. 2C (a power capacity of 1 MW) does not contain SC and FES because they are not adequately rated to supply this amount of power. Fig. 2D for a power capacity of 10 MW does not show SC, SMES, or FES for the same reason. Although most technologies have their LCOE begin to level out by 50 MWh of storage capacity, it should be noted that CAES continues to decrease as storage capacity increases. Therefore, CAES has a competitive LCOE at large storage capacities.

4. Comparison of ES technologies

In this section, the tabulated values of ES devices will be displayed graphically to create useful comparisons to aid an industrial facility's DSM efforts. It is important to note the tabulated values displayed here are achieved only under ideal conditions. For example, SC ES, SMES, and FES are only highly efficient when the storage time is short because they all tend to have high self-discharge rates. As another example, the power density of Li-ion BES is only in this range when it is operated at its normal temperature range due to impacts on the chemical reactions within the battery.

First, Fig. 3 compares the efficiency and power density of all ES devices. This is an important comparison for a facility to estimate the amount of power that can be extracted from a device for the space that is available, and the efficiency of the device. Clearly a more efficient technology will yield more cost savings since less energy will be needed to yield the same amount of output energy after storage. As shown from Fig. 3, it is no surprise that the ES technologies that directly store electrical energy (i.e., SC and SMES) yield the highest efficiencies. Also, FES shows many similar characteristics to SC ES or SMES, so they are often combined for readability. It is also seen that CAES has the lowest efficiency when compared to the rest of the technologies. This occurs because when air is compressed to high pressures (increasing potential energy) there is a large amount of associated waste heat and this potential energy must be converted back to electrical energy.

The next comparison, seen in Fig. 4, compares the lifetime to the power density of the ES devices. Power density is selected as the independent variable in Figs. 3 and 4 because a facility will likely be selecting an ES device based on both available space and demand reduction required. Fig. 4 is a useful comparison for selection of an ES device because the user can select a device based on the desired demand reduction, space available, and the duration of time that the device will be operational, some of the most important qualities for an industrial facility to consider in their DSM efforts. The trend again shows SC ES and SMES as technologies with the longest lifetimes with high power densities, but it should be kept in mind that these technologies may not be well suited for use in industrial facilities because they cannot store a large amount of energy.

Next, the discharge time is compared to the power density for each ES device in Fig. 5. This is arguably the most meaningful comparison when matching an ES device to a facility because both the discharge time and power density must meet the needs of the facility for it to be a useful match. If the ES device cannot discharge for the duration needed to shave a facility's peak load, at the power capacity required, in the available space, it is useless to the facility. Although SC ES, SMES, and FES have very high efficiencies, it is seen that they also have the lowest discharge times. Therefore, these technologies may not be useful to a facility if they cannot discharge for a long enough time to shave the peak demand, despite their high efficiencies. Conversely, even though CAES had a low efficiency, it has one of the longest discharge times along with PHES.

In the next plot, Fig. 6, the energy density is compared to the specific energy. This plot shows that CAES stores more energy per unit kilogram than per unit liter, which makes sense because its density is low compared to the other technologies. It also shows that Li-ion and NaS BES systems have the highest values of both energy density and specific energy, whereas PHES and SMES have the lowest.

Comparing the time dependence of ES technologies can be another useful way to match an ES technology to an industrial facility for DSM efforts. The response time is plotted against the discharge time in Fig. 7. In order to adequately manage demand, an ES device will need to discharge long enough to ensure that a facility's demand will be reduced, but it also must respond quickly enough to shave an unexpected peak load. BES seems to be the best option, as it can both respond quickly and discharge for a long duration. SC and SMES fall short in their discharge time, even though they respond extremely fast.

In the next plot, Fig. 8, the capital cost is compared to the rated power to cross compare between technical characteristics and costs. This plot can be used by industry to assess the facility's budget to the required magnitude of peak shaving needed from an ES device. Since cost is generally the driving factor for DSM, comparing the up-front (capital) costs to the power that will be supplied can be very meaningful. This figure should be used in combination with Fig. 5 to match an energy

Table 5
LCOE and Lifetime of ES Devices ($/kWh).

	T (years)	100 kW/100 kWh	100 kW/400 kWh	500 kW/2 MWh	1 MW/4 MWh	10 MW/40 MWh
Li-ion	10	6.99	2.11	0.95	0.81	0.68
NaS	13	7.59	2.05	0.80	0.64	0.50
Pb-acid	10	4.11	1.27	0.71	0.64	0.58
Flow	8	3.68	1.01	0.38	0.30	0.23
SC	30	0.008	–	–	–	–
SMES	25	0.14	0.13	–	–	–
FES	15	0.012	–	–	–	–
PHES	50	1855.87	463.97	92.85	46.46	4.71
CAES	30	260.77	65.20	13.08	6.57	0.70

Note '-' indicates that the device cannot meet the power and/or energy storage capacity.

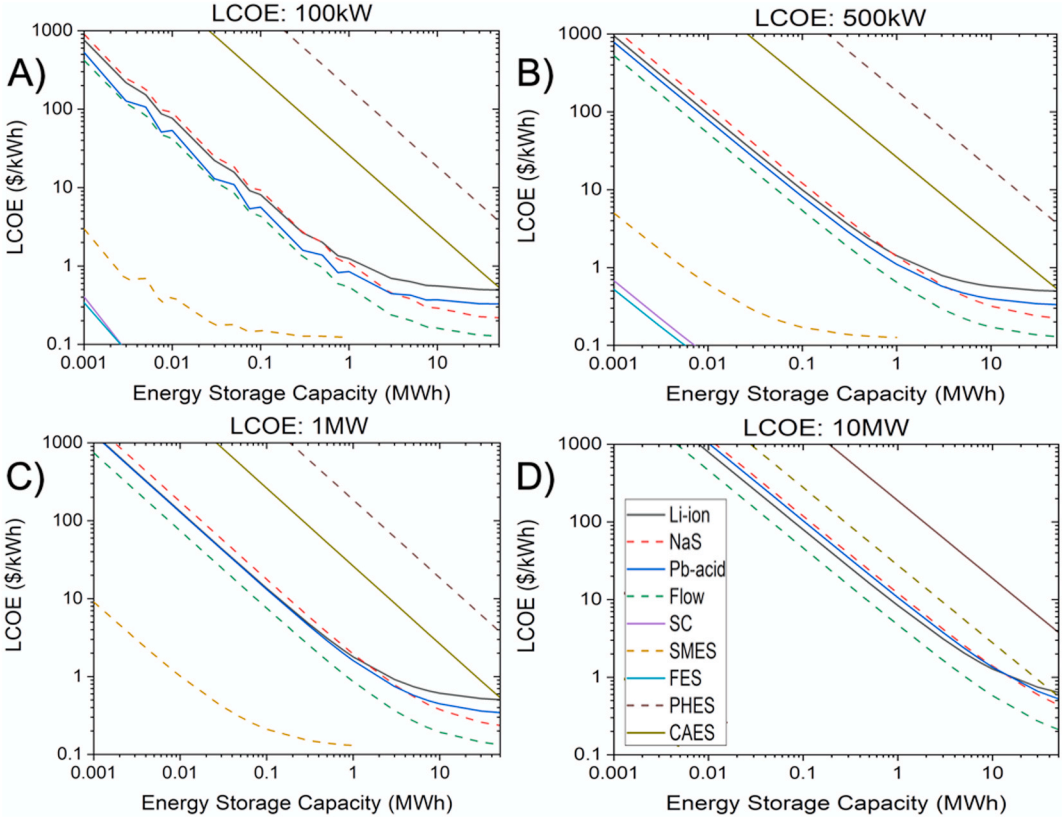


Fig. 2. LCOE of ES devices as functions of power and energy storage capacity.

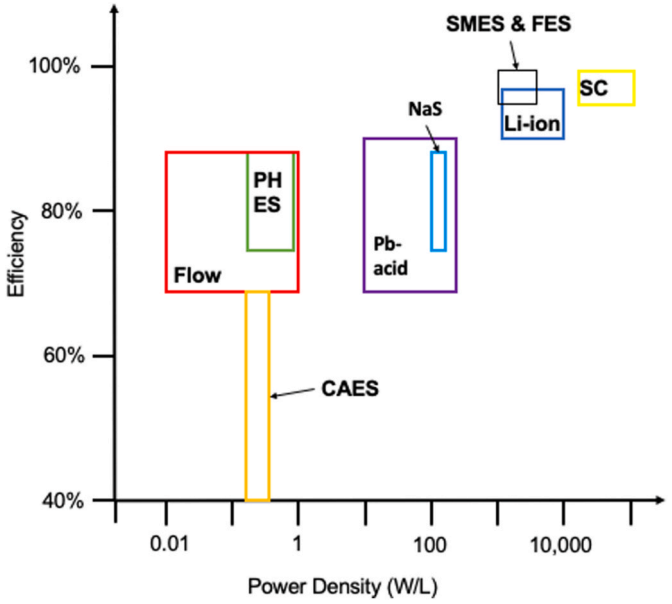


Fig. 3. Efficiency vs. Power Density of ES Devices.

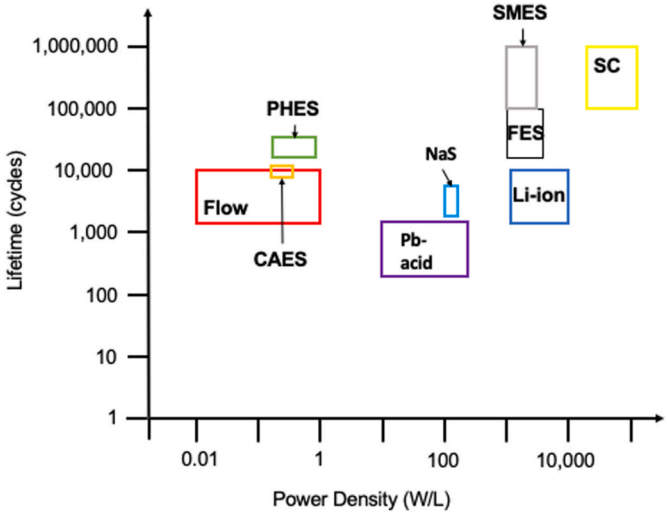


Fig. 4. Lifetime vs. Power Density of ES Devices.

storage technology to a facility. To explain, a facility will first decide upon a technology that can achieve the power rating needed to shave their peak demand by a certain value using the x-axis of Fig. 8. The rated power will be divided by the available space in the facility for energy storage to determine the power density required from the device (x-axis of Fig. 5). The suitable technologies are now filtered to those that meet the rated power condition and power density condition. Then, the y-axes of Figs. 5 and 8 will be used to ensure the discharge time and capital costs requirements of the facility are satisfied.

In the next plot, Fig. 9, the lifetime is compared to the capital costs. This is an important comparison for a facility to assess how long their ES device will last, based on their capital cost budget. Using Figs. 8 and 9 together can give a facility a reasonable estimate about capital costs relationship to power output and lifetime. It is again seen that SC ES and SMES are the best technologies on this plot because of their low capital costs and long lifetimes, but the deterrents of short discharge times and low rated powers must be remembered. However, it can also be observed that the large capital costs of PHES, Li-ion BES, and Flow BES are justified by their long lifetimes.

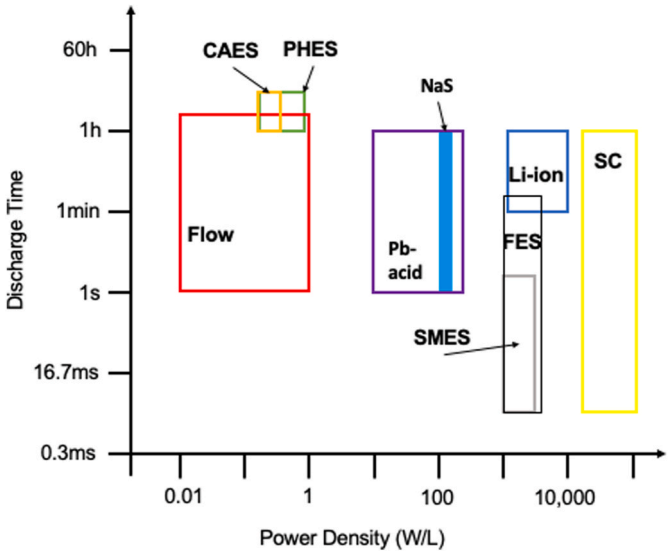


Fig. 5. Discharge Time vs. Power Density of ES Devices.

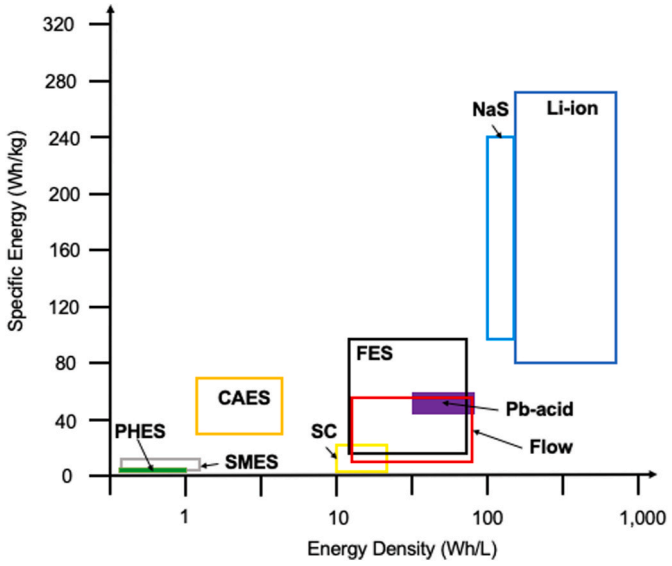


Fig. 6. Specific Energy vs. Energy Density of ES Devices.

In the next plot, Fig. 10, the O&M costs are compared to the capital costs. Here it is observed that Li-ion BES has high capital costs and O&M costs, whereas SC ES and SMES have low capital costs and low O&M costs. Of the BES technologies, Pb-acid BES has the lowest O&M costs and similar capital costs to the other BES technologies. Furthermore, although PHES has high capital costs, its O&M costs are low making it inexpensive to operate once the system is purchased.

A useful way to compare ES devices is based on their power capacity. Recall the common power and ES capacities for DSM in industrial facilities were 100 kW/100 kWh, 100 kW/400 kWh, 500 kW/2 MWh, 1 MW/4 MWh, and 10 MW/40 MWh. The capacities considered here are strictly power capacity ranges of 0–500 kW, 500 kW – 5 MW, and 5–50 MW because a facility will likely choose a device based on the demand reduction needed. However, while power capacity is important, other properties are also critical to a facility's DSM efforts like power density, efficiency, lifetime, discharge time, response time, and costs. Therefore, Table 6 is organized in such a fashion to see the best three devices for a power range and a selected property shown in Table 1, Table 2, and Table 3. As an example, in the power range 500 kW–5 MW, Li-ion BES

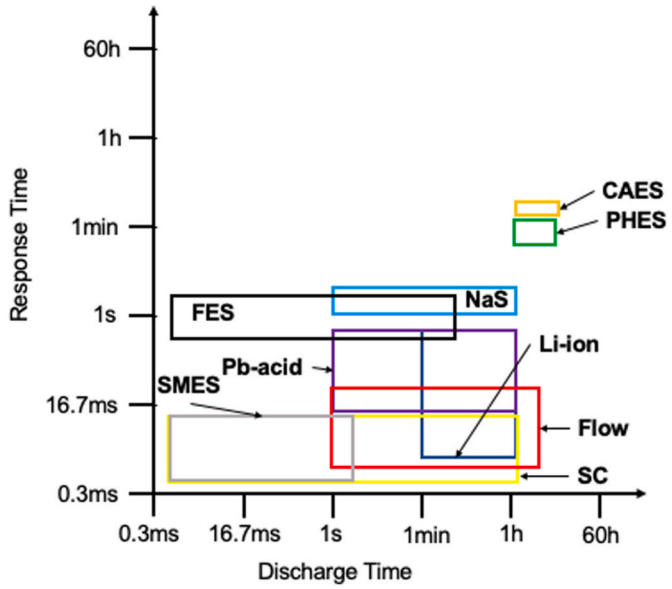


Fig. 7. Response Time vs. Discharge Time of ES Devices.

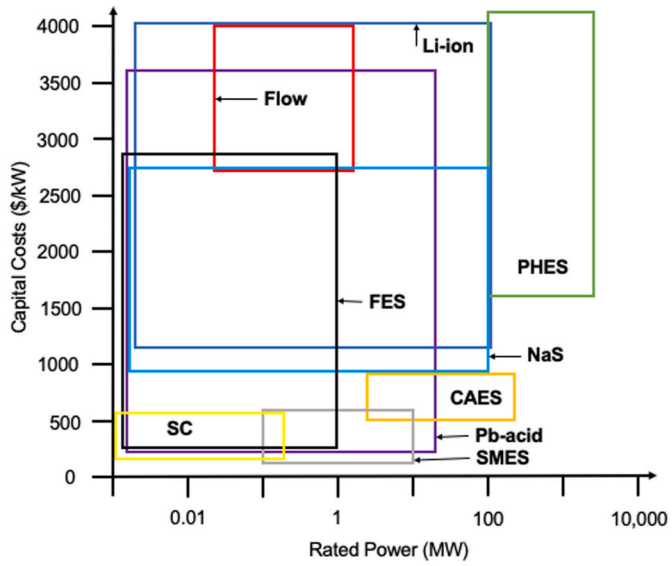


Fig. 8. Capital Costs vs. Rated Power of ES Devices.

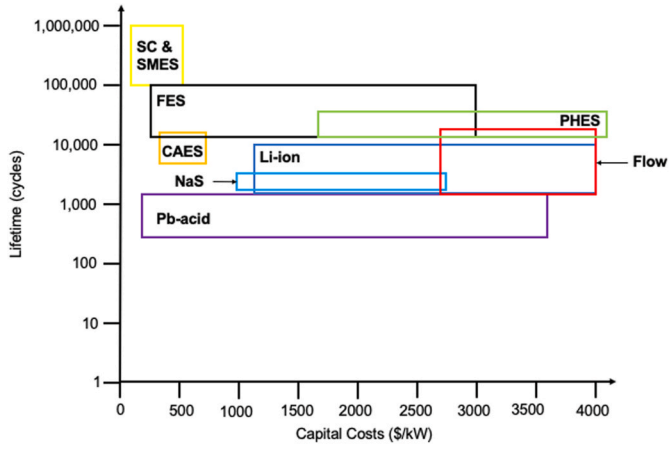


Fig. 9. Lifetime vs. Capital Costs of ES Devices.

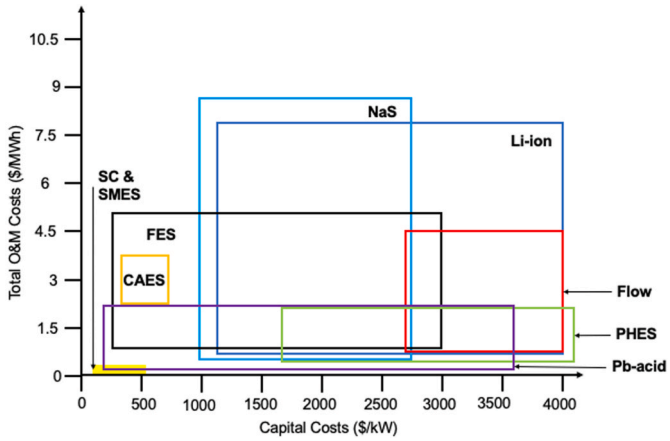


Fig. 10. Total O&M Costs vs. Capital Costs of ES Devices.

Table 6
Top 3 ES Devices based on Power Capacity.

	0–500 kW	500kW-5MW	5MW–50MW
Specific Energy	NaS	NaS	NaS
	Li-ion	Li-ion	Li-ion
	CAES	CAES	CAES
Energy Density	Li-ion	Li-ion	Li-ion
	NaS	NaS	NaS
	FES	FES	Pb-acid
Specific Power	SC	SMES	Li-ion
	SMES	FES	Pb-acid
	FES	Li-ion	NaS
Power Density	SC	Li-ion	Li-ion
	Li-ion	FES	Pb-acid
	FES	SMES	NaS
Efficiency	SMES	SMES	Li-ion
	SC	Li-ion	NaS
	Li-ion	FES	Pb-acid
Lifetime	FES	FES	PHES
	SC	SMES	CAES
	SMES	PHES	Flow
Storage Time	Flow	Flow	Flow
	PHES	PHES	PHES
	CAES	CAES	CAES
Response Time	SC	SMES	Flow
	SMES	Flow	Pb-acid
	Flow	FES	Li-ion
Discharge Time	CAES	CAES	CAES
	PHES	PHES	PHES
	Flow	Flow	Flow
Capital Cost	SC	SMES	CAES
	SMES	CAES	Pb-acid
	CAES	FES	PHES
Capital Storage	PHES	PHES	PHES
	CAES	CAES	CAES
	NaS	NaS	NaS
Fixed O&M	CAES	CAES	CAES
	FES	FES	Flow
	Flow	Flow	Li-ion
Variable O&M	Pb-acid	Pb-acid	Pb-acid
	PHES	PHES	PHES
	Flow	Flow	Flow
Total O&M	PHES	PHES	PHES
	Pb-acid	Pb-acid	Pb-acid
	Flow	Flow	Flow

has the highest power density, followed by FES and SMES. Although SC ES has a higher power density than Li-ion BES, it cannot output power at a rate greater than 500 kW. Also, it should be noted that even though the highest power density is considered the "best," the lowest value of response time, and other properties that should be minimized, are considered the "best." An example of this is in the 0–500 kW range, where SC ES has the fastest response time, followed by SMES and flow BES.

Technologies that appear many times in the table above can be considered a "well-rounded" technology since they possess numerous favorable characteristics. Therefore, Table 7 summarizes the number of times an ES technology appears in the table above, for each power range.

Upon observation of Table 7, it can be seen that in the power range of 0–500 kW, SC ES, flow BES, and CAES most often have desirable properties, whereas between 500 kW and 5 MW, Pb-acid BES, Flow BES, and CAES most often have desirable properties. In the range of 5–50 MW, Pb-acid BES is ranked as a "top 3" technology for 12 different properties. Overall, CAES is ranked in the "top 3" technologies the most times, followed by Pb-acid BES, Li-ion BES, and PHES.

However, this comparison is including only the values in Table 1, Table 2, and Table 3, but a capital cost per cycle and LCOE were also derived for typical power and ES capacities of industrial facilities. Recall for the capacities 100 kW/100 kWh, SC ES has the lowest capital cost per cycle and LCOE, but for all other capacities PHES has the lowest capital cost per cycle. When considering LCOE values, SMES has the lowest for 100 kW/400 kWh, and flow BES has the lowest for all other capacities. Although CAES is ranked as a "top 3" technology the most times, its LCOE is highest for the capacity of 1 MW/4 MWh. Although flow BES is only rated as a "top 3" technology 14 times, it has the lowest LCOE for most of the power and ES capacities shown. Pb-acid BES is ranked as a "top 3" technology 21 times and does not have the highest LCOE value for any of the power and ES capacities, but does have the highest capital cost per cycle for 100 kW/100 kWh. Li-ion BES is rated as a "top 3" technology 19 times and does not have the largest LCOE for any power and ES capacity, but it does have the largest capital costs per cycle for every power and ES capacity except for 100 kW/100 kWh.

It must be remembered that although SC, SMES, and FES have suitable technical characteristics, their discharge time is likely not long enough for use in many industrial facilities for DSM. The 100 kW/100 kWh power and ES capacity was included to compare these ES technologies against the others, but this power and ES capacity resulted in some cost anomalies. For example, this range is the only one that Pb-acid BES had the highest capital cost per cycle, and Li-ion BES had the lowest.

When picking "well-rounded" technologies, they should have several occurrences in Table 6, and have a low capital cost per cycle and LCOE. Therefore, based on the analysis above, "well-rounded" technologies can be considered as CAES, Pb-acid BES, Li-ion BES, PHES, and Flow BES.

5. Case studies

Based on the database (National Technology and Engineering Solutions of Sandia (NTESS), 2020), any facilities that use ES for demand response are grouped by their broadly defined industry to recognize trends in selected ES technologies. The broadly defined industries include industrial, commercial, educational institutions, and public buildings. Trends in the installations of the technology used, i.e. number of installations, are compared to the total rated power capacity of the technology installed. For example, of the 23 industrial facilities found using ES for demand response, 6 are using NaS batteries (or 26.09% of the installations) at an installed capacity of 25 MW (46.1% of the 54.2 MW of power capacity across all 23 facilities). It should be noted that

Table 7
Occurrences of technologies in Table 6.

Technology	0–500 kW	500 kW-5 MW	5–50 MW	Total
Li-ion	4	6	9	19
NaS	3	4	0	7
Pb-acid	2	7	12	21
Flow	6	7	1	14
SC	6	0	0	6
SMES	5	6	0	11
FES	5	0	0	5
PHES	5	5	9	19
CAES	6	7	11	24

refrigeration capacities were divided by an assumed average coefficient of performance of 4 to convert from refrigeration capacity to electrical power capacity. Based on the analysis above, it can be expected that the "well-rounded" technologies of Li-ion BES, Pb-acid BES, flow BES, PHES, and CAES will have numerous applications, likely to a high capacity. Even though CAES and PHES typically take advantage of large geographical conditions (elevation changes, reservoirs, and caverns), there is potential for small-scale applications within these technologies.

Based on the results shown in Fig. 11, it is seen that industrial facilities ES trends are different than that of commercial facilities, educational institutions, and public buildings. It should be noted that industrial facilities primarily consist of manufacturing and production facilities of any kind, and water reclamation centers. There are several water reclamation centers in Japan that use NaS BES on a large scale, hence the large power capacity. Li-ion BES has the greatest number of installations, to no surprise since it was deemed a "well-rounded" technology. Commercial facilities are primarily that of office buildings and corporate headquarters, so the results make sense that TES is installed at many facilities due to large spaces that need either cooling or heating. One somewhat surprising result is that flow BES is installed at the largest total capacity for commercial facilities. This is surprising because the technology is not well known, but it is not surprising because it was considered a "well-rounded" technology in the analysis above. Flow BES also has a large number of installations for industrial facilities, but the total capacity remains low. For educational institutions, primarily colleges and universities, it is expected that TES is installed with the highest frequency and capacity. Public buildings consist of hospitals, airports, community centers, and other buildings openly available to the general public, and have Li-ion BES installed most frequently, with TES installed to the largest capacity.

In all, the case studies primarily yield results that are expected, but also some unexpected results. It is seen that "well-rounded" technologies, like Li-ion BES are often used across all types of facilities. There are also installations of Pb-acid BES and flow BES scattered across different facilities, and it is surprising that the "well-rounded" technologies of PHES and CAES are not installed at any facilities, even on a small scale. Since there are small-scale applications of these ES technologies that perform similar to large-scale applications, it is a surprise that CAES and PHES are not yet common technologies for DSM in industrial facilities. However, since other considerations must be made to install these technologies, like locating an elevation change or suitable pressure vessel, these technologies can be viewed as unattractive. Also, cost drawbacks of these technologies include high capital costs to install PHES and additional costs throughout the project's lifetime resulting from the low efficiency of CAES.

6. Conclusions

The aim of this review was to determine well rounded energy storage technologies for use in industrial DSM. The analyses conducted herein

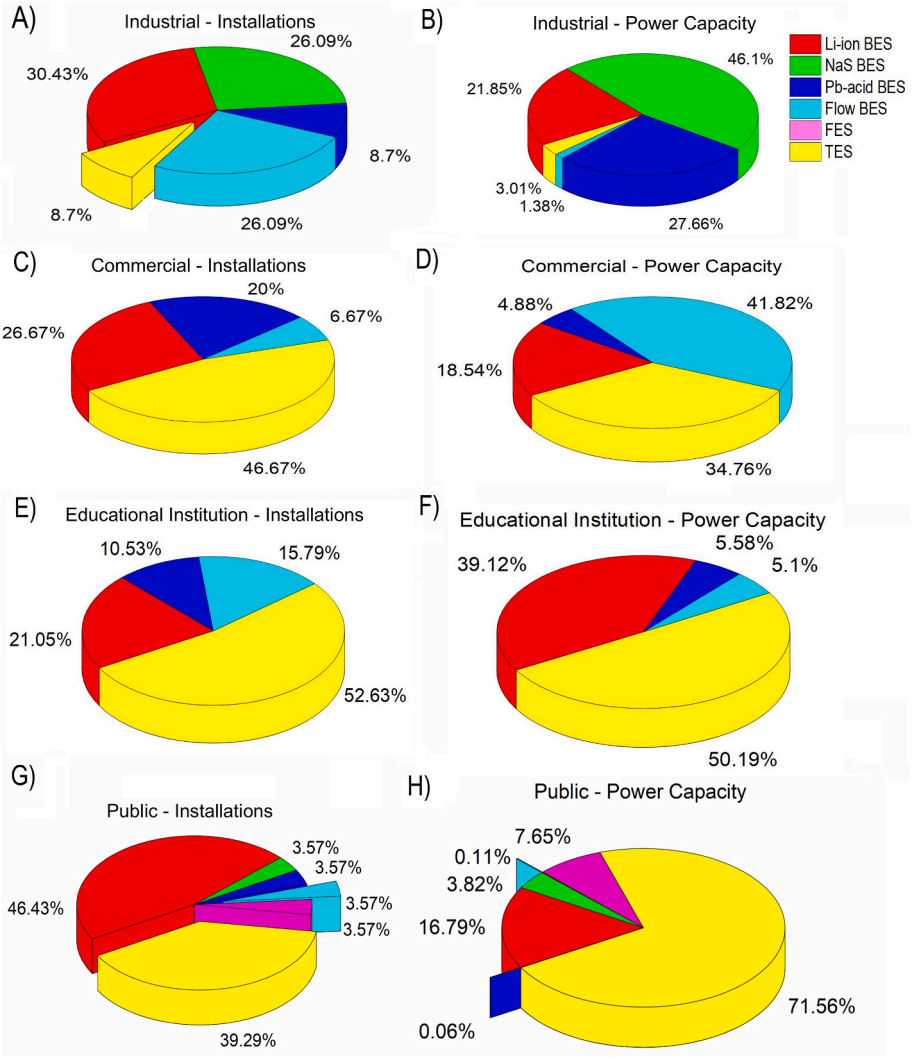


Fig. 11. Case study results.

deemed Li-ion BES, Pb-acid BES, flow BES, PHES, and CAES as "well-rounded" technologies, meaning that they perform well across all power capacities and most properties discussed. Although they are "well-rounded," each technology features a crucial disadvantage. For example, Li-ion BES is the most expensive ES technology reviewed, Pb-acid BES has the shortest lifetime, flow BES has the lowest power density, PHES is typically installed only on a very large scale, and CAES is the least efficient.

To better understand the different applications in industrial facilities, numerous case studies are conducted to discover which ES technologies are applied to certain industries for DSM. Pertaining to industrial facilities, Li-ion BES is the most often installed ES for DSM, with NaS BES having the greatest capacity. Commercial facilities and educational institutions have TES most commonly installed, while educational institutions and public buildings have TES installed to the greatest capacity. Since industrial facilities are unique facilities having high electricity usage for production, their choice in ES is seen to be drastically different than other facilities analyzed, typically storing electrical energy rather than thermal energy. To explain, since their heating and air conditioning loads are generally not as high as facilities that are designed for high occupancy, TES is installed less often.

Although "well-rounded" ES devices are determined herein, this does not mean that these are optimal for every facility. Many parameters must be evaluated when matching an ES technology to a facility. For example, load curves, stochasticity of the peak (magnitude and time of day), demand vs. energy costs, and amount of demand reduction desired. Most importantly, the price that each facility is willing to pay for energy storage depends on their capital, electricity rates, production/cost, and revenue structures. Ultimately, an optimization model to match an ES device to a facility must be created to further this work in applying ES for industrial DSM.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jclepro.2021.127322.

Nomenclature

A-CAES	Adiabatic Compressed Air Energy Storage
BES	Battery Energy Storage
CAES	Compressed Air Energy Storage
CPR-CAES	Constant Pressure Ratio Compressed Air Energy Storage
D-CAES	Diabatic Compressed Air Energy Storage
DSM	Demand-Side Management
ES	Energy Storage
FES	Flywheel Energy Storage
I-CAES	Isothermal Compressed Air Energy Storage
IWH	Industrial Waste Heat
LCOE	Levelized Cost of Energy
Li-ion	Lithium Ion
MES	Mechanical Energy Storage
NaS	Sodium Sulfur
Pb-acid	Lead Acid
PCM	Phase Change Material
PHES	Pumped Hydro Energy Storage
PSB	Polysulphide Bromide Battery
RES	Renewable Energy System
RFB	Redox Flow Battery
SC	Supercapacitor
SMES	Superconducting Magnetic Energy Storage
TES	Thermal Energy Storage
VPR-CAES	Variable Pressure Ratio Compressed Air Energy Storage
VRB	Vanadium Redox Battery
ZnBr	Zinc Bromide

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